# 2.3 Symmetric and asymmetric encryption

## 2.3.1 The purpose of encryption

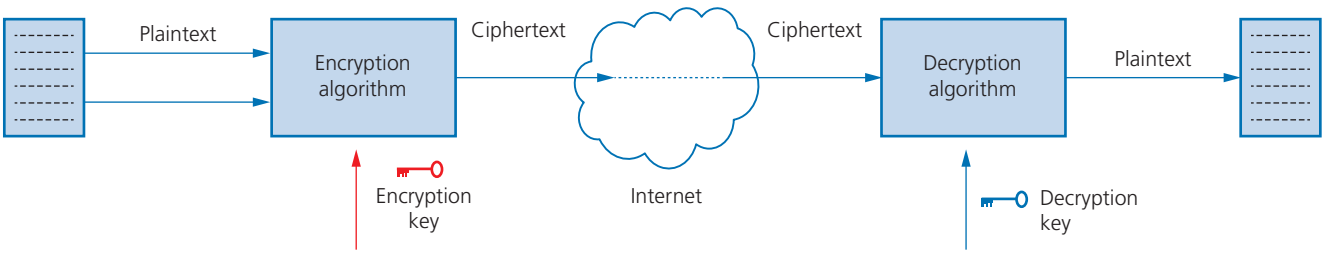
When data is transmitted over any public network (wired or wireless), there is always a risk of it being intercepted by, for example, a hacker. Under these circumstances, a hacker is often referred to as an **eavesdropper**. Using **encryption** helps to minimise this risk.

> **Link**
>
> For more on cyber security see Chapter 5.

Encryption alters data into a form that is unreadable by anybody for whom the data is not intended. It cannot prevent the data being intercepted, but it stops it from making any sense to the eavesdropper. This is particularly important if the data is sensitive or confidential (for example, credit card/bank details, medical history or legal documents).

### Plaintext and ciphertext

The original data being sent is known as **plaintext**. Once it has gone through an **encryption algorithm**, it produces **ciphertext**:




▲ **Figure 2.18** Plaintext and ciphertext

## 2.3.2 Symmetric and asymmetric encryption

### Symmetric encryption

**Symmetric encryption** uses an encryption key; the same key is used to encrypt and decrypt the encoded message. First of all, consider a simple system that uses a 10-digit denary encryption key (this gives $1 \times 10^{10}$ possible codes); and a decryption key. Suppose our encryption key is:

4 2 9 1 3 6 2 8 5 6

which means every letter in a word is shifted across the alphabet +4, +2, +9, +1, and so on, places. For example, here is the message COMPUTER SCIENCE IS EXCITING (plaintext on the top line of Figure 2.19) before and after applying the encryption key (forming the ciphertext shown on the bottom line of Figure 2.19):

| C | O | M | P | U | T | E | R |
|---|---|---|---|---|---|---|---|
| 4 | 2 | 9 | 1 | 3 | 6 | 2 | 8 |
| G | Q | V | Q | X | Z | G | Z |

| S | C | I | E | N | C | E |
|---|---|---|---|---|---|---|
| 5 | 6 | 4 | 2 | 9 | 1 | 3 |
| X | I | M | G | W | D | H |

| I | S |
|---|---|
| 6 | 2 |
| O | U |

| E | X | C | I | T | I | N | G |
|---|---|---|---|---|---|---|---|
| 8 | 5 | 6 | 4 | 2 | 9 | 1 | 3 |
| M | C | I | M | V | R | O | J |

▲ **Figure 2.19** Plaintext into ciphertext using 10-digit encryption key

To get back to the original message, it will be necessary to apply the same decryption key; that is, 4 2 9 1 3 6 2 8 5 6. But in this case, the decryption process would be the reverse of encryption and each letter would be shifted –4, –2, –9, –1, and so on. For example, 'G' → 'C', 'Q' → 'O', 'V' → 'M', 'Q' → 'P', and so on.

However, modern computers could 'crack' this encryption key in a matter of seconds. To try to combat this, we now use 256-bit binary encryption keys that give $2^{256}$ (approximately, $1.2 \times 10^{77}$) possible combinations. (Even this may not be enough as we head towards **quantum computers**.)

**Find out more**

One of the ways of mitigating the risk of symmetric keys falling into the wrong hands (known as the key distribution problem) is to use a system based on modulo-11, where both sender and receiver can calculate the encryption key without it actually being exchanged in any way. Find out how this system works.

The real difficulty is keeping the encryption key a secret (for example, it needs to be sent in an email or a text message which can be intercepted). Therefore, the issue of security is always the main drawback of symmetrical encryption, since a single encryption key is required for both sender and recipient.

## Asymmetric encryption

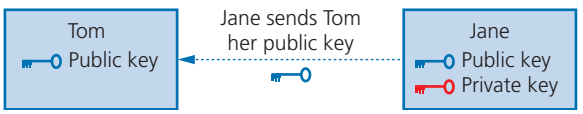
**Asymmetric encryption** was developed to overcome the security problems associated with symmetric encryption. It makes use of two keys called the **public key** and the **private key**:

- public key (made available to everybody)
- private key (only known to the computer user).

Both types of key are needed to encrypt and decrypt messages.

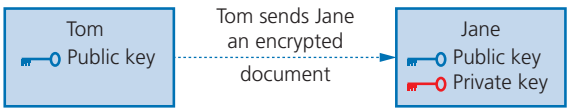
We will use an example to explain how this works; suppose Tom and Jane work for the same company and Tom wishes to send a confidential document to Jane:

1. Jane uses an algorithm to generate a **matching pair of keys** (private and public) that they must keep stored on their computers; the matching pairs of keys are mathematically linked but can't be derived from each other.
2. Jane now sends her public key to Tom.




▲ **Figure 2.20** Jane sends Tom her public key

3. Tom now uses Jane's public key (🔑) to encrypt the document he wishes to send to her. He then sends his encrypted document (ciphertext) back to Jane.




▲ **Figure 2.21** Encrypted document sent from Tom to Jane

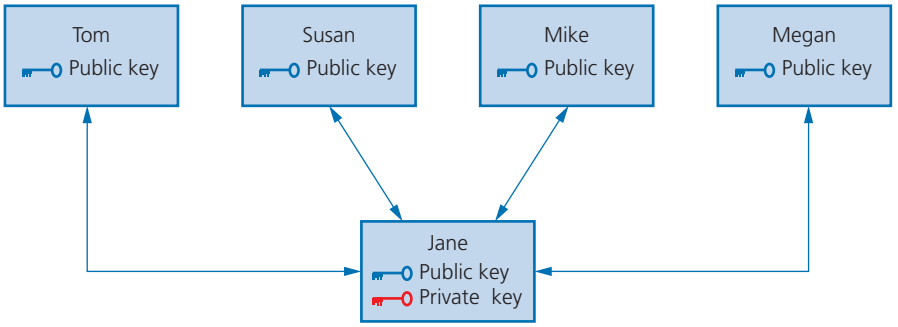
4. Jane uses her **matching** private key (🔑) to unlock Tom's document and decrypt it; this works because the public key used to encrypt the document and the private key used to decrypt it are a matching pair generated on Jane's computer. (Jane can't use the public key to decrypt the message.)

## Activity 2.8

1 At the moment Jane can only receive encrypted documents from Tom.
Describe what would need to happen for Jane to be able to **send** encrypted documents back to Tom.

2 Explain why this method is much more secure than symmetric encryption.

Jane can also exchange her public key with any number of people working in the company, so she is able to receive encrypted messages (which have been encrypted using her public key) and she can then decrypt them using her matching private key:




▲ **Figure 2.22** The sharing of Jane's public key

However, if a two-way communication is required between all five workers, then they all need to generate their own matching public and private keys. Once this is done, all users then need to swap public keys so that they can send encrypted documents/files/messages between each other. Each worker will then use their own private key to decrypt information being sent to them.

### Find out more

1 Using Figure 2.22 as your template, draw a new diagram showing the public keys and private keys that need to be swapped if Jane wishes to have a two-way exchange of encrypted documents between Tom, Susan, Mike and Megan.

2 Consider the complexity, if all five people want to have secure two-way communication between each other (and not just with Jane). This would mean each of the five workers sharing their own public keys with each of the other workers.

## Activity 2.9

For each of the following ten questions, choose which of the five options corresponds to the correct response.

a What is meant by the term **ciphertext** when used in encryption?

**A** an encryption or decryption algorithm

**B** an encrypted message

**C** a type of session key

**D** another name for plaintext

**E** text following an encryption algorithm

**b** When carrying out asymmetric encryption, which of the following users would keep the private key?

**A** the sender
**B** the receiver
**C** both sender and receiver
**D** all recipients of the message
**E** none of the above

**c** In encryption, which of the following is the term used to describe the message before it is encrypted?

**A** simpletext
**B** plaintext
**C** notext
**D** ciphertext
**E** firsttext

**d** Which of the following is the biggest disadvantage of using symmetric encryption?

**A** it is very complex and time consuming
**B** it is rarely used anymore
**C** the value of the key reads the same in both directions
**D** it only works on computers with older operating systems
**E** there is a security problem when transmitting the encryption key

**e** Which of the following is the correct name for a form of encryption in which both the sender and the recipient use the same key to encrypt and decrypt?

**A** symmetric key encryption
**B** asymmetric key encryption
**C** public key encryption
**D** same key encryption
**E** block cipher encryption

**f** What of the following is the final number in a code, which is calculated from all the numbers in the code; its purpose is to find errors in data entry?

**A** parity check
**B** checksum
**C** cyclic redundancy check
**D** parity bit
**E** check digit

**g** Which of the following is a form of error detection that makes use of a system of acknowledgements and timeouts?

**A** automatic repeat request
**B** echo check
**C** check digit
**D** parity bit
**E** cyclic redundancy check

**h** Which of the following methods uses an extra bit added to a byte to ensure it contains an even number of 1s or odd number of 1s?

**A** cyclic redundancy check
**B** parity check
**C** checksum
**D** check digit
**E** echo check

**i** Which of the following uses a calculated value which is sent after a block of data; the receiving computer also calculates the value from the block of data and compares the values?

**A** parity check
**B** check digit
**C** packet switching
**D** checksum
**E** automatic repeat request

j Which of the following describes the check where the receiving computer sends back a copy of the data to the sending computer to allow it to compare the data?

**A** echo check
**B** automatic repeat request
**C** checksum
**D** parity check
**E** check digit

## Extension

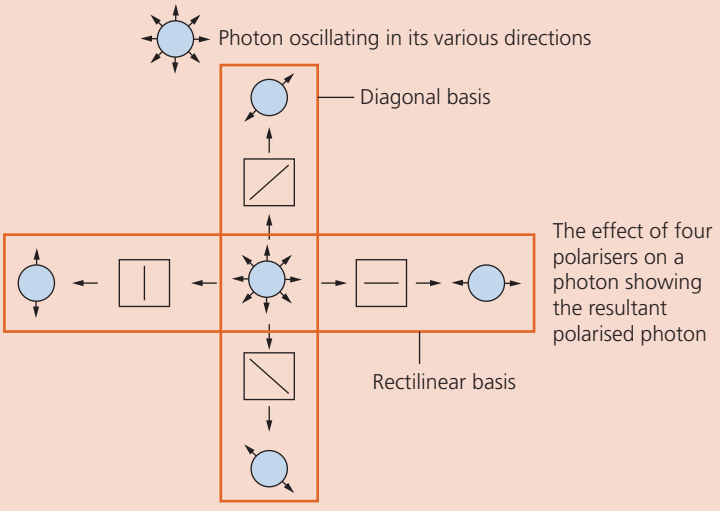
For those students considering the study of this subject at A Level, the following section gives some insight into further study on encryption.

**Quantum cryptography** exploits the laws of quantum mechanics to improve on the security of data. Quantum cryptography is based on the use of particles of light called photons (with energy calculated according to the formula: E = hf) and their physical quantum properties to produce a virtually unbreakable encryption system. This helps protect the security of data being transmitted over fibre optic cables. The technology is based on the laws of physics, rather than mathematics which is how the current cryptography methods already covered in this chapter work.

One of the uses of quantum cryptography is when sending encryption keys across a network – this uses a **Quantum Key Distribution (QKD)** protocol (one of the most common is called BB84).

QKD uses quantum mechanics to ensure a secure transmission of encryption keys. They use a **Qubit** (**Qu**antum **bit**) which is the basic unit of quantum 'data'. Unlike normal binary (which uses discrete 0s and 1s), the state of a Qubit is **both** 0 and 1 until the photon is measured.

A photon normally vibrates or oscillates in all different directions. Polarisation restricts these vibrations to particular directions. The following diagram shows how a photon can be polarised in one of two **bases** – the rectilinear (up/down and side-to-side) basis and the diagonal basis. Do not worry too much about what a basis is – you can just think of them as two different ways of preparing and measuring the photon.




▲ **Figure 2.23** Quantum cryptography

So how do we use quantum cryptography to send an encryption key from 'A' to 'B' using the QKD protocol? To understand this we need to note that:

- Photons can be polarised in one of two bases – rectilinear or diagonal.
- In the rectilinear basis, 1 can be represented by ↕ and 0 by ↔. In the diagonal basis 1 is ↗ and 0 is ↖.
- A photon that is polarised in the rectilinear basis will always give the same result when measured in the rectilinear basis. So, if a photon is polarised as '1' in the rectilinear basis (↕) it will always be measured in the rectilinear basis to have the polarisation ↕.
- Similarly, a photon that is polarised in the diagonal basis will always give the same result when measured in the diagonal basis. So, if a photon is polarised as '1' in the diagonal basis (↗) it will always be measured in the diagonal basis to have the polarisation ↗.
- However, if a photon is polarised in the rectilinear basis but measured in the diagonal basis then information about the original polarisation is lost. The result of the measurement has a 50-50 chance of being ↗ or ↖ because in the diagonal basis the photon is in both 1 and 0 states **at the same time!** This measurement tells the receiver nothing about the photon's original polarisation in the rectilinear basis.
- Similarly, if a photon is polarised in the diagonal basis but measured in the rectilinear basis then information about the original polarisation is lost. The result of the measurement has a 50-50 chance of being ↕ or ↔ because in the rectilinear basis the photon is in both 1 and 0 states **at the same time**! This tells the receiver nothing about the photon's original polarisation in the diagonal basis.

1. So, the sender polarises each photon using a basis that is selected at random.
2. The receiver measures each photon using a basis that is selected at random.
3. The sender and receiver publicly exchange which basis they used for each photon.

Only when the sender and receiver use the same basis for measurement can they be sure that are both reading a 1 or 0. When they used the same basis the receiver knows they have measured a '1' or '0' correctly.

In this chapter, you have learnt about:

- ✔ the use of data packets when sending data over networks
- ✔ packet switching
- ✔ types of data transmission (serial, parallel, simplex, half-duplex and full-duplex)
- ✔ the universal serial bus (USB)
- ✔ how errors during transmission can occur and how to recognise them and recover from them (using parity check, checksum and echo check)
- ✔ the use of check digits to identify errors following data entry
- ✔ how automatic repeat requests are used to find errors in transmitted data
- ✔ why encryption of data is used
- ✔ symmetric and asymmetric encryption
- ✔ the use of public keys and private keys in asymmetric encryption.